
UNIFIED MECHANICS

Volume III

Traversal, Time, Observation, and Readout

Joseph Shields

VOLUME III OF V

Scientific Edition · 2 August 2026

A CORRIGIBLE EDITION. This text is corrigible. Earlier manifests and printings are retained as provenance, not as immutable authority. Corrections are recorded by version and supersession history; no scientific claim is frozen.

ABSTRACT

This volume follows a distinction through propagation, recurrence, measurement, record formation, causal order, and observer-dependent readout. It integrates the traversal papers, helical-relativity programme, Universopedia, and the field-assembly and lossless-transcription results.

Contents

1	TRAVERSAL DEPTH	1
1.1	Recurrence is not return of the complete state	1
1.2	The helix as a normal form	1
1.3	Topology does not supply an arrow	2
2	CAUSAL ORDER AND CLOSED TIMELIKE CURVES	3
2.1	What causal reconstruction establishes	3
2.2	Gödel spacetime as a boundary case	3
2.3	Consistency, memory, and identity	4
3	HOLOGRAPHIC AND SCALE LIFTS	5
3.1	Radial evolution and renormalization-group flow	5
3.2	Finite domains and hidden history	5
4	FIELD ASSEMBLY AND LOSSLESS TRANSCRIPTION	7
4.1	Assembly is not a list of parts	7
4.2	The evolution-preserving transcription criterion	8
4.3	Connection to gauge transport	8
5	MECHANICAL OBSERVATION	9
5.1	The complete measurement chain	9
5.2	Five ledger types	9
5.3	Actual boundaries are channel-dependent	9
5.4	Observer architecture	10
6	TIME, ENSEMBLES, AND COMPLEXITY	11
6.1	Four distinct clocks	11
6.2	When a collection becomes a field	11
6.3	A vector, not a single complexity essence	12
6.4	The periodic table as a complete language family	12
7	TRAVERSAL TESTS AND HONEST LIMITS	13
7.1	A retained-fraction threshold	13
7.2	What the traversal programme establishes	13

7.3 Forward link 14

Traversal depth

1.1 Recurrence is not return of the complete state

Many problems in the corpus share one mathematical pattern: a projection makes an evolution look closed while a variable omitted by the projection continues to change. Let

$$\pi : \tilde{X} \rightarrow X \quad (1.1)$$

forget a history coordinate. A sequence may satisfy $x_{n+p} = x_n$ in X while its lift satisfies

$$\tilde{x}_{n+p} = T\tilde{x}_n, \quad T \neq I. \quad (1.2)$$

The projected orbit is periodic; the full orbit carries monodromy. A circle is the projection of a helix when one coordinate is suppressed.

The corpus calls the omitted coordinate *traversal depth*. The name is useful if it is treated operationally: depth is whatever state variable distinguishes repeated projected positions that have different prior histories or future responses. Merely appending an integer counter always removes projected recurrence mathematically. The physical question is whether the counter is co-variant, dynamically coupled, and measurable.

INTERNAL RESULT 1.1 — LIFTED NONCLOSURE

Let $h : \tilde{X} \rightarrow \mathbb{R}$ satisfy $h(T\tilde{x}) = h(\tilde{x}) + \Delta h$ with $\Delta h > 0$. Then no orbit with nonzero winding under T is closed in $\tilde{X}$, even if its projection is closed in X . This is exact mathematics of the chosen lift; it does not establish that nature carries h .

1.2 The helix as a normal form

For a projected angular variable $\theta \sim \theta + 2\pi$, a simple lift is

$$\gamma(s) = (R \cos \omega s, R \sin \omega s, vs). \quad (1.3)$$

After one projected period $2\pi/\omega$, the first two coordinates recur and the third advances by $2\pi v/\omega$. The helix is therefore a visualization of Equation (1.2), not

a claim that every worldline is literally a corkscrew in an additional spatial dimension.

The helical-relativity documents extend this picture to clock order, holographic scale, phase, and record history. The scientifically safe unification is:

A recurrent base description may require a nonrecurrent lifted variable whenever the omitted state changes the result of a later admissible operation.

That statement is just the harmful-compression criterion of Volume I applied to recurrence.

1.3 Topology does not supply an arrow

A universal cover distinguishes windings, but topology alone does not privilege increasing over decreasing winding number. Orientation and monotonicity require dynamics, boundary conditions, or a record criterion. Consequently, the corpus's arrow cannot be obtained by drawing a helix. The helix represents an arrow after a monotone history variable has been specified.

EXPOSURE 1.1 — THE MISSING PHYSICAL LIFT

The traversal programme still owes a covariant definition of depth, an evolution equation for it, a coupling to ordinary fields, and an observable that changes when the depth changes. Without those elements, the lift is a faithful bookkeeping extension but not an additional physical degree of freedom.

Causal order and closed timelike curves

2.1 What causal reconstruction establishes

In Minkowski spacetime of dimension greater than two, suitable bijections preserving causal relations are generated by Lorentz transformations together with translations and dilations under the hypotheses of Alexandrov–Zeeman-type results (Zeeman, 1964; Alexandrov, 1967). For sufficiently well-behaved relativistic spacetimes, the family of timelike curves determines conformal and topological structure under distinguishing conditions, as in Malament’s theorem (Malament, 1977).

ESTABLISHED RESULT 2.1 — CAUSAL STRUCTURE IS GEOMETRICALLY POWERFUL

Causal order can determine the conformal metric structure of a spacetime under specific dimensional, regularity, bijectivity, and causal-distinguishing hypotheses. It does not determine an absolute conformal scale, and the theorem cannot be applied when its causal hypotheses fail.

This reverses a common intuition: geometry is not always prior to causal order. Nevertheless, it does not show that every metric must be derived from the corpus’s record order. It shows that a precisely defined causal relation carries substantial geometric information.

2.2 Gödel spacetime as a boundary case

Gödel’s rotating cosmological solution of Einstein’s equations contains closed timelike curves (Gödel, 1949). It therefore fails standard chronology and distinguishing conditions needed by causal reconstruction. This makes it a useful stress test for any claim that an observer’s ordered record is fundamental.

The corpus proposes that a closed timelike curve in the base spacetime lifts to an open curve in record space. The proposal is consistent as an augmentation: the traveller returns to a spacetime event while a history coordinate has

advanced. It is not a theorem of general relativity, because general relativity's state is whatever the model includes. Adding the coordinate changes the model.

2.3 Consistency, memory, and identity

Deutsch formulated quantum consistency conditions near closed timelike curves using fixed-point density operators (Deutsch, 1991). Gavassino analysed a closed system on a closed timelike curve and found that global consistency can require recurrence of the complete internal state, including erasure of memories accumulated around the loop (Gavassino, 2024).

The corpus responds that a state whose new record is erased is not a continuation of the same record-bearing identity. This is a legitimate semantic criterion, but it does not invalidate the consistency solution. It changes what the word “same observer” is required to preserve.

CONDITIONAL INTERPRETATION 2.1 — IDENTITY ACROSS A CLOSED TIMELIKE CURVE

If observer identity is defined by persistent recoverability of accumulated records, then a globally consistent loop that erases those records does not return the same record-bearing observer in that sense. This is an identity criterion, not a proof that the spacetime solution is dynamically forbidden.

The inherited “no recollapse while the record stands” claim is likewise not an established result of gravitation. General relativity admits expanding and recollapsing solutions. A no-overwrite law would have to appear as a covariant constraint or stress-energy contribution and survive cosmological tests.

Holographic and scale lifts

3.1 Radial evolution and renormalization-group flow

The AdS/CFT correspondence relates a gravitational theory in an asymptotically anti-de Sitter bulk to a conformal field theory on its boundary in its best-understood settings (Maldacena, 1998; Witten, 1998). The holographic radial coordinate is connected to scale, and Hamilton–Jacobi equations for the bulk action can be written in renormalization-group form (de Boer et al., 2000).

The corpus observes that some conventions express boundary-to-bulk or decreasing-scale evolution with a minus sign. Replacing that sign by a positive “completed history” variable is a reparameterization when the map is monotone. A geometric lift carries more information only if two states at the same projected scale can have distinct histories that later matter.

Let s be a flow parameter and $W(s)$ a monotone functional with

$$\frac{dW}{ds} \leq 0. \quad (3.1)$$

Define accumulated evolution

$$E(s) = W(0) - W(s) \geq 0. \quad (3.2)$$

Equation (3.2) is an exact change of variables. It does not add a dimension unless E is retained independently of W and affects observables.

3.2 Finite domains and hidden history

In a finite or periodic base domain, a scale or radial coordinate can recur or saturate while the sequence of coarse-graining operations remains different. A lifted state

$$\tilde{x} = (x, E) \quad (3.3)$$

retains that sequence. This is useful computationally for non-Markovian processes: the enlarged state can make an apparently history-dependent evolution Markovian. Again, the enlargement is justified only when it improves prediction of a declared target.

RESEARCH TARGET 3.1 — PHYSICAL HOLOGRAPHIC DEPTH

Identify a holographic observable or response function that depends on accumulated traversal history at fixed conventional radial data, and derive the history rate from the bulk/boundary dynamics rather than choosing it. Absent such an observable, the lift is explanatory notation.

Field assembly and lossless transcription

4.1 Assembly is not a list of parts

The corpus uses deliberately artificial examples—including the “Crowley” and “Glorpnorp” constructions—to demonstrate a general point: a collection of component descriptions does not determine the assembled system unless incidence, order, orientation, boundary conditions, and coupling rules are also transmitted.

Formally, let components be $c_1, \dots, c_n$ and let R denote assembly relations. A source system is

$$X = (\{c_i\}, R). \quad (4.1)$$

A transcript retaining only $\{c_i\}$ identifies all systems with the same parts, regardless of R . If the evolution operator F depends on R , no decoder can reconstruct $F(X)$ from the part list alone.

INTERNAL RESULT 4.1 — FIELD-ASSEMBLY FAILURE

Let $C(\{c_i\}, R) = \{c_i\}$. If there exist $R \neq R'$ and a relevant evolution/readout Q for which

$$Q(\{c_i\}, R) \neq Q(\{c_i\}, R'), \quad (4.2)$$

then Q does not factor through C . No assembly procedure using only the component list can be lossless for Q .

This theorem is elementary but widely applicable. A field catalogue without spatial adjacency cannot reproduce a gradient-dependent dynamics. A Lagrangian’s bulk equation without boundary data cannot reproduce a boundary charge. A periodic coordinate without winding cannot reproduce history-dependent response.

4.2 The evolution-preserving transcription criterion

Let $T : X \rightarrow Z$ be a transcript and F_X, F_Z the source and transcript evolutions. The transcript is dynamically sufficient when the diagram commutes:

$$T \circ F_X = F_Z \circ T. \quad (4.3)$$

If only a readout R must be preserved, the weaker target-relative requirement is

$$R_X \circ F_X = R_Z \circ F_Z \circ T. \quad (4.4)$$

The first criterion supports autonomous simulation in the transcript space; the second supports a specific observable.

The corpus's Principle of Logical Action is most rigorous when read as Equation (4.3): an action representation is admissible for a task only when its compression commutes with the relevant evolution and readout. It is not necessary to claim that every physical law is literally linguistic.

4.3 Connection to gauge transport

Comparing fields at separated points requires a rule of transport. A gauge connection supplies precisely that rule; curvature records path-dependent non-closure. Thus “relation must propagate” has a standard mathematical realization in gauge geometry. The corpus's new claim is not the existence of connections, but the attempt to derive their necessity from lossless comparison. Completing that derivation requires specifying the local equivalence group and proving that any admissible comparison rule is a connection on the corresponding bundle.

Mechanical observation

5.1 The complete measurement chain

A physical description becomes an empirical statement only through a typed chain:

$$\begin{aligned} \text{mechanism} &\rightarrow \text{state variable} \rightarrow \text{observable} \rightarrow \text{coupling} \\ &\rightarrow \text{record} \rightarrow \text{calibration} \rightarrow \text{inference.} \end{aligned} \quad (5.1)$$

Every arrow can discard information or introduce model dependence. The Universopedia layer of the corpus is strongest when used as an audit of Equation (5.1), not as a new force law.

Let X_S be the world-state space at scale S , $D_{S,c}$ the record space of channel c , and

$$M_{S,c} : X_S \rightarrow D_{S,c} \quad (5.2)$$

the complete measurement map, including propagation, coupling, selection, instrument response, and noise. A decoder assigns a record to a validated language class, while an inference model estimates physical parameters. These maps must not be conflated.

5.2 Five ledger types

The difference between T and F is crucial. A feature in the transmission kernel cannot be recovered from the same record by a cleverer classifier. A feature in the familiarity gap is present but not understood; it may become classifiable without new data.

5.3 Actual boundaries are channel-dependent

The “actual Boundary” is defined as the causal cut separating retained identity from exchanged information at a selected scale, channel, and timescale. It is not necessarily a material surface. A star has different optical, neutrino, acoustic, gravitational, nuclear, and thermodynamic boundaries. A black-hole event horizon, a photosphere, and a detector threshold are different cuts.

TABLE 5.1. The observational ledger.

Code	Type	Meaning
R	calibrated record	stored count, image, spectrum, time series, strain, track, or catalogue value
E	established inference	model-dependent physical estimate with units, uncertainty, and validity range
U	framework reconstruction	Unified Mechanics interpretation layered on R and E
T	transmission limit	distinctions the chosen channel did not preserve
F	familiarity gap	structure present in the record but not yet assigned to a validated class

The Sun illustrates the point without requiring a new theory. Photons generated in the interior undergo radiative transport and repeated interaction before escape, whereas neutrinos weakly couple and provide a much more direct channel from nuclear reactions in the core. Solar-neutrino experiments have measured both flux and flavour transformation (SNO Collaboration, 2002; Borexino Collaboration, 2018). The same source can therefore be opaque to one carrier and comparatively transparent to another. Opacity is a property of matter, carrier, energy, and channel together.

5.4 Observer architecture

A mechanical observer is represented by six stages:

1. prepared ready state;
2. channel-defining coupling;
3. propagation through the apparatus;
4. physical resolution or correlation;
5. amplification into robust degrees of freedom;
6. production and retention of a calibrated record.

Human semantic interpretation may follow, but it is not required for physical observation. This preserves the corpus's broad observer definition while keeping measurement dynamics separate from consciousness.

Time, ensembles, and complexity

6.1 Four distinct clocks

Age is not one scalar unless an identity class and clock are specified. Univerpedia records

$$\tau = (\tau_{\text{signal}}, \tau_{\text{formation}}, \tau_{\text{material}}, \tau_{\text{processing}}). \quad (6.1)$$

Signal age

propagation delay from emission or event to the record.

Formation age

time since assembly of the selected effective object.

Material age

distribution of ages of constituent matter or nuclei.

Processing age

time since the last substantial local reconfiguration.

The four clocks can differ by orders of magnitude. A present-day planetary atmosphere contains nuclei older than the planet, has weather processed on short timescales, and is observed with a finite light-travel delay. Collapsing these into “the age” erases distinctions relevant to different questions.

6.2 When a collection becomes a field

For samples (x_i, z_i) and a scale-dependent kernel W_ℓ , define

$$F_\ell(x) = \frac{\sum_i W_\ell(x - x_i) f(z_i)}{\sum_i W_\ell(x - x_i)}, \quad N_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \quad (6.2)$$

When $N_{\text{eff}} \approx 1$, the output remains effectively a local-object value. Field language becomes operationally justified when many contributions are involved, the result is stable under defensible changes of resolution, and the continuum approximation predicts the target observables within uncertainty.

6.3 A vector, not a single complexity essence

The corpus defines a scale-relative profile

$$\kappa = (D, N, R, C, M, U), \quad (6.3)$$

for diversity, nesting, retention, channels, model depth, and uncertainty. The profile is descriptive. Averaging ordinal components can assist navigation but is not a law of nature, and uncertainty must not be hidden inside a single rank.

6.4 The periodic table as a complete language family

The periodic table is the corpus's principal example of a mature observational language: atomic identity is indexed by nuclear charge Z , refined by isotope A , ionization, electronic state, and environment. Spectral lines, isotope shifts, selection rules, and mass records allow the same atomic classes to be recognized across laboratories, stars, interstellar media, and engineered systems.

This does not make the periodic table a derivation from Unified Mechanics. It demonstrates what a successful familiarity system looks like:

1. stable class invariants;
2. multiple independent measurement channels;
3. environment-dependent subclasses;
4. explicit transmission limits such as opacity, blending, resolution, and selection rules;
5. conservative extension when a new isotope, state, or line class is learned.

The complete element-by-element ledger is preserved in Volume V's source and data crosswalk rather than repeated as foundational physics.

Traversal tests and honest limits

7.1 A retained-fraction threshold

The corpus proposes r^3 as the loss assigned to a three-stage traversal. A possible optical test converts it into a retained fidelity threshold

$$\mathcal{F} \geq 1 - r^3 \approx 0.9704915028. \quad (7.1)$$

If expressed through a declared optical metric such as Strehl ratio, the metric and acceptance rule must be fixed before data are examined. The number is a forecast only after the physical mapping from traversal stages to the measured system is fixed.

FORECAST PROGRAMME 8.1 — PARAMETER-FREE TRAVERSAL RETENTION

Choose a physical channel with exactly three independently justified stages, specify a state-distance or wavefront metric, register Equation (7.1) and all analysis choices, and test on withheld data. A stage count selected after observing the loss would make the result a postdiction.

7.2 What the traversal programme establishes

TABLE 7.1. Traversal and observation ledger.

Statement	Grade	Scientific boundary
Projected recurrence can hide state change	exact mathe- matics	relevance depends on future observables
Helical lift removes base recurrence	exact construc- tion	physical depth is not yet derived
Causal order constrains conformal geometry	established	requires causal and regularity hypotheses
Gödel spacetime contains closed timelike curves	established	not experimentally realised by the corpus

Statement	Grade	Scientific boundary
Memory-erasing recurrence breaks record-defined identity	conditional interpretation	does not invalidate the space-time solution
Holographic scale can be written as Hamilton–Jacobi flow	established in its domain	new history coordinate remains conjectural
Assembly requires relations as well as components	exact factorisation result	target-relative
Observation is mechanism/channel/record/inference chain	methodological standard	each application needs a concrete model
Transmission and familiarity kernels differ r^3 traversal threshold	exact conceptual distinction forecast proposal	familiarity is classifier-relative needs preregistered stage and metric map

7.3 Forward link

Volume IV applies these rules to the corpus’s physical claims. It reports established group-theoretic reproductions, finite sampling results, cosmological comparisons, simulation programmes, and null tests. Every empirical section begins with the measurement map and ends with the unresolved kernel.

Bibliography

- A. D. Alexandrov. A contribution to chronogeometry. *Canadian Journal of Mathematics*, 19:1119–1128, 1967. doi: 10.4153/CJM-1967-102-6.
- Borexino Collaboration. Comprehensive measurement of pp -chain solar neutrinos. *Nature*, 562:505–510, 2018. doi: 10.1038/s41586-018-0624-y.
- Jan de Boer, Erik Verlinde, and Herman Verlinde. On the holographic renormalization group. *Journal of High Energy Physics*, 2000(08):003, 2000. doi: 10.1088/1126-6708/2000/08/003.
- David Deutsch. Quantum mechanics near closed timelike lines. *Physical Review D*, 44:3197–3217, 1991. doi: 10.1103/PhysRevD.44.3197.
- Lorenzo Gavassino. Life on a closed timelike curve. *Classical and Quantum Gravity*, 42(1):015002, 2024. doi: 10.1088/1361-6382/ad98df.
- Kurt Gödel. An example of a new type of cosmological solutions of einstein’s field equations of gravitation. *Reviews of Modern Physics*, 21:447–450, 1949. doi: 10.1103/RevModPhys.21.447.
- David B. Malament. The class of continuous timelike curves determines the topology of spacetime. *Journal of Mathematical Physics*, 18:1399–1404, 1977. doi: 10.1063/1.523436.
- Juan M. Maldacena. The large- N limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2:231–252, 1998. doi: 10.1023/A:1026654312961.
- SNO Collaboration. Direct evidence for neutrino flavor transformation from neutral-current interactions in the sudbury neutrino observatory. *Physical Review Letters*, 89:011301, 2002. doi: 10.1103/PhysRevLett.89.011301.
- Edward Witten. Anti-de sitter space and holography. *Advances in Theoretical and Mathematical Physics*, 2:253–291, 1998. doi: 10.4310/ATMP.1998.v2.n2.a2.
- E. C. Zeeman. Causality implies the lorentz group. *Journal of Mathematical Physics*, 5:490–493, 1964. doi: 10.1063/1.1704140.